Information

Lab 7 Exercise: Views and User-Defined Functions

STRICT INSTRUCTIONS:

The function and view names must exactly match those given in the question.

Strictly follow: When defining functions, set the delimiter to $$ before the function definition and reset it to; after the definition.

DO NOT CHANGE OR REMOVE THE QUERIES GIVEN IN THE TEMPLATE FILE FOR 1b, 2b, 3b, 4b, AND 5c

Q1: 1 mark

a) Write a user-defined function named calculate_bonus. The bonus is 20% of salary for Managers and 10% for all the other roles.

This function accepts the emp_id as input and returns the calculated bonus amount.

b) Execute the select query provided in the template.

Expected output:

```
+---------------------+
| calculate_bonus(109) |
+---------------------+
|             9000.00 |
+---------------------+
1 row in set (0.01 sec)
```

Q2: 1 mark

a) Create a view named it_staff that displays all details of employees who work in the 'IT' department (where dept_id = 1).

b) Execute the select query provided in the template to display the contents of this view.

Expected output:

```
+--------+---------------+-----------+----------+---------+
| emp_id | emp_name      | job_role  | salary   | dept_id |
+--------+---------------+-----------+----------+---------+
|    101 | Alice Johnson | Developer | 75000.00 |       1 |
|    102 | Bob Smith     | Manager   | 95000.00 |       1 |
|    105 | Eve Black     | Developer | 72000.00 |       1 |
+--------+---------------+-----------+----------+---------+
3 rows in set (0.00 sec)
```

Q3: 1 mark

a) Create a view named emp_locations that joins the Employees and Departments tables.

The view should contain: emp_id, emp_name, dept_name, and location.

b) Execute the select query provided in the template to display the contents of this view.

Expected output:

```
+--------+---------------+------------------+---------------+
| emp_id | emp_name      | dept_name        | location      |
+--------+---------------+------------------+---------------+
|    101 | Alice Johnson | IT               | San Francisco |
|    102 | Bob Smith     | IT               | San Francisco |
|    105 | Eve Black     | IT               | San Francisco |
|    103 | Charlie Brown | HR               | New York      |
|    104 | David White   | Sales            | Chicago       |
|    106 | Frank Green   | Marketing        | Los Angeles   |
|    107 | Grace Hall    | Finance          | Boston        |
|    108 | Henry King    | Engineering      | Seattle       |
|    109 | Ivy Lee       | Customer Support | Austin        |
|    110 | Jack Scott    | Operations       | Denver        |
+--------+---------------+------------------+---------------+
10 rows in set (0.01 sec)
```

Q4: 1 mark

a) Create a user-defined function get_tax_bracket.

This function takes emp_id as input and returns a varchar string based on the salary.

if salary >= 90000: return ‘High’

if salary between 60000 and 89999: return ‘Medium’

if salary below 60000: return ‘Low’

b) Execute the select query provided in the template.

Expected output:

```
+----------------------+
| get_tax_bracket(104) |
+----------------------+
| Medium               |
+----------------------+
1 row in set (0.00 sec)
```

Q5: 2 marks

a) Create a view named low_salary_group. The view must include all columns from the employees table, and the relevant records for the corresponding tax bracket.

b) Write a query to give a bonus to the employees who come under the "Low" tax bracket using the view and function created in the previous questions.

c) Execute the select query provided in the template.

Expected output:

```
+--------+----------+
| emp_id | salary   |
+--------+----------+
|    109 | 54000.00 |
+--------+----------+
1 row in set (0.00 sec)
```

◄ Lab 7 zip file

Jump to...

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